

SIMATS ENGINEERING

ASSESSMENT TOOL 1 (High)- Analytical Problem Solving

COURSE NAME: Data Structures

COURSE CODE:CSA0316

COURSE OUTCOME COVERED

CO2 : Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. (BTL3, BTL4)

Assessment Tool Weightage: **30%**

QUESTIONS:

1. A queue-based ticketing system serves customers in the order of arrival, but VIP customers must be prioritized occasionally. Analyze how modifying a simple queue into a priority-based queue affects fairness, waiting time, and computational complexity. Compare both systems using analytical reasoning.

Queue-Based Ticketing System:

Simple Queue vs Priority Queue.

A queue-based ticketing system is commonly used in banks, hospitals, railway stations, and customer service centers to serve customers in an organized manner. A simple queue follows the FIFO (First In, First Out) principle, where the customer who arrives first is served first.

However, in some situation, VIP customers or emergency cases need to be served before others. In such cases, the system is modified into a priority queue, where customers are served based on their priority rather than their arrival time.

Fairness

A simple queue is considered the fairest system because every customer is treated equally. No customer is allowed to skip the queue, so everyone waits according to their arrival order. This ensures transparency and customer satisfaction.

In contrast, a priority queue gives preference to VIP or high priority customers. While this improves service for important customers, it reduces fairness because regular customers may have to wait longer even if they arrived earlier.

Waiting Time:-

In a simple queue, the waiting time depends only on the number of customers already in the queue. Every customer knows that they will be served after those who arrived before them, making the waiting time predictable.

In a priority queue, waiting time is reduced for VIP customers because they are served immediately or before regular customers. However, the waiting time for normal customers increases since they can be delayed by higher-priority customers. Therefore, the average waiting time becomes unequal among customers.

Computational Complexity:-

A simple queue is easy to implement using an array or Linked List. The enqueue (insert) and dequeue (delete) operations both take $O(1)$ time, making it efficient for large numbers of customers.

A priority queue is usually implemented using a heap. In this case, inserting a customer and removing the highest priority customer require $O(\log n)$ time because the priority order must be maintained. Although this is slightly slower than a simple queue, it provides better service for high-priority customers.

<u>Comparison</u>		
Feature	Simple Queue	Priority Queue.
Service order	First In, First out	Based on customer priority
Fairness	High	Lower, as VIP's are prioritized
Waiting Time	Equal and predictable	Less for VIP's, more for regular customers
Enqueue operation	$O(1)$	$O(\log n)$
Dequeue operation	$O(1)$	$O(\log n)$
Applications	Banks, ticket counters, billing systems	Hospitals, airports, emergency services, VIP ticketing.

Conclusion:

A simple queue is the best choice when fairness and equal treatment of customers are the main objectives. A priority queue is more suitable when certain customers, such as VIP's or emergency cases, require faster service. Although a priority queue increases computational complexity and may reduce fairness, it provides better service equality for high-priority customers.

2. Customers are standing in a line before a Banker Cashier Counter to perform bank transactions. What data structure will be suitable to perform customer transactions. Give a suitable data structure diagram to serve the customer without bias.

2. Customers standing in a Line Before a Bank Cashier Counter

The most suitable data structure for saving customers at a bank cashier counter is a Queue. A queue follows the FIFO principle, which means the customer who arrives first is served first. This ensures fairness, avoids bias, and maintains an orderly flow of customer transactions.

Working of Queue in a Bank.

The queue operations in the following steps:-

- * A customer enters the bank and joins the end of the queue.
- * The customer is added to the queue using the enqueue operation.
- * The cashier calls the customer standing at the front of the queue.
- * The transaction (deposit, withdrawal, etc) is completed.
- * The customer leaves the queue after the transaction.
- * The next customer automatically moves to the front and is served.
- * This process continues until the queue becomes empty.

Queue Operations

* Enqueue.

→ Add a new customer to the rear of the queue.

* Dequeue.

→ Removes the customer from the front after completing the transaction.

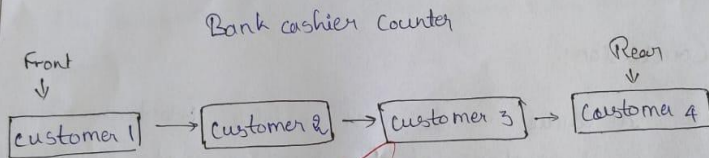
* Peek.

→ Display the customer currently waiting at the front without removing them.

* IsEmpty

→ checks whether there are any customers waiting in the queue.

Queue Data Structure Diagram.



Cashier serves customer 1 first

followed by customer 2, customer 3, & customer 4.

Advantages of Using Queue.

- * Ensures fair service by following the FIFO principle.
- * Prevents customers from skipping the line.
- * Maintains discipline and transparency.
- * Simple to implement and manage.
- * Efficient with $O(1)$ insertion and deletion operations.
- * Reduces confusion and improves customer satisfaction.

Time Complexity

Operation	Time Complexity
Enqueue	$O(1)$
Dequeue	$O(1)$
Peek	$O(1)$
IsEmpty	$O(1)$

Conclusion:-

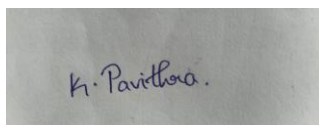
A Queue is the most suitable datastructure for customer transactions at a bank cashier counter because it serves customers in the exact order of their arrival. It ensures fairness, prevents bias, maintains an organized flow of customers, and provides efficient transaction processing. Due to its simplicity, efficiency and transparency the queue is widely used in banks and other customer service systems.

Code of Conduct:

I K. Pavithra (192472155) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

**RUBRICS FOR CALCULATION:**

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation